Project Proposal

infrared.city-1: Tree detection

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1 Project Description

This chapter outlines the motivation, methods used, data sources and the scope of this project.

1.1 Motivation and Project Overview

As urban areas are expanding and global temperatures rise due to climate change, being able to understand how trees can help cool cities has become increasingly important [1]. Furthermore, trees reduce the risk of flooding in cities and promote health and wellbeing [2]. Our project aims at taking the first step in the direction of better tree coverage in urban areas by creating a model which is able to predict tree locations using satellite imagery of densely populated areas.

1.2 Data Sources

For our project we use publicly available Sentinel-2 satellite images with 10-meter spatial resolution and four color channels, namely red, green, blue and near infrared. The images are retrieved via the API of the Copernicus Programme and are provided by the European Space Agency [3]. As seasonality has a great impact on the appearance of trees, imagery of different seasons will be used to train the model. Additionally, we will use the publicly available data set on every tree in the city of Vienna "Baumkataster" to train and test our model on exact tree locations [4].

1.3 Model

Former papers suggest that for image segmentation with satellite data the U-Net deep learning model is the most accurate [5] and will therefore also be used in this paper to predict tree locations. It is a special type of Convolutional Neural Network and consists of a contracting path and an expanding path. In the contracting path convolution and pooling layers are used to extract features from the original imagery, which in this project will consist of four pictures for each season and four color channels each as described above. The expanding path consists of the same structure of convolution and pooling layers but used in the other direction to up-sample the before contracted imagery [6]. As suggested in similar work this project will use skip connections between the expanding and contracting path as well as a temporal attention algorithm to prioritize more meaningful images [7]. The Binary Cross-Entropy loss between the predicted locations will be used and the actual locations from the “Baumkataster” will be split into a train and test set.

1.4 Project Scope

The model in this paper is trained only on satellite images of urban areas and therefore also only supposed to be used for tree detection within city environments. Also, the model focuses on tree detection in the temperate climate zone as this is the vegetation it is trained to observe. Further analysis on the impact of trees on, for example, heat regulation in cities and flooding prevention are out of the scope of this project and could be the topic of future research.

2 BEAM

The figure below presents the BEAM representation [8] of the proposed tree detection system, illustrating the data sources, training and inference processes, model architecture, evaluation pipeline, and associated risks.

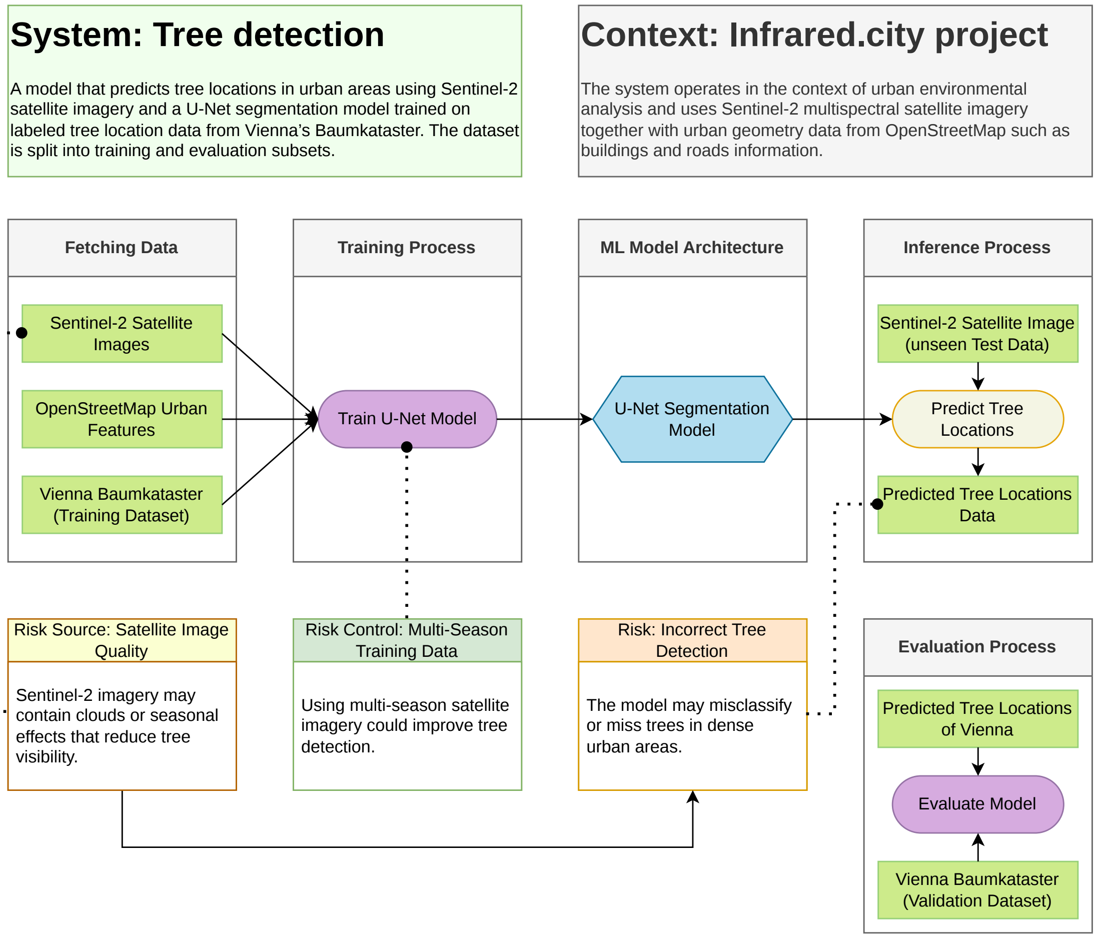

Figure 1: BEAM model of the proposed tree detection system.

3 Timeplan

The project will be conducted between March and June and is structured into several phases corresponding to the course milestones and deliverables.

3.1 Phase 1: Project Setup & Planning
March 9 – March 19

This initial phase focuses on defining the project scope, exploring available data sources, and establishing a solid project plan.

Tasks:
- Kick-off session and first meeting with data coaches (March 9)
- Define project objective and research question
- Explore available datasets (Sentinel-2, OpenStreetMap, etc.)
- Decide on modelling approach
- Assign team roles
- Draft project plan

Deliverable: Project Plan – March 19

3.2 Phase 2: Methodology Development

March 20 – March 29

During this phase, the primary goal is to acquire the necessary data and establish the foundational methodology for feature extraction.

Tasks:
- Data collection and preprocessing
- Feature extraction (spectral indices, vegetation signals)
- Define modelling pipeline
- Improve project plan based on feedback

Deliverable: Refined Project Plan – March 29

3.3 Phase 3: Data Processing & Initial Modelling

March 30 – April 21

This period is dedicated to executing the data processing pipeline and training the first iteration of machine learning models.

Tasks:
- Implement data processing pipeline
- Train initial machine learning models
- Evaluate early results
- Prepare intermediate analysis
- Create presentation slides

Deliverable: Intermediate Report and Slides – April 21

3.4 Phase 4: Model Improvement & Validation

April 22 – June 6

Focus now shifts towards optimizing model performance through rigorous validation, error analysis, and peer feedback.

Tasks:
- Intermediate consultations (April 22–24)
- Improve model performance
- Perform validation and error analysis
- Conduct sparring group meetings
- Prepare final report draft

Deliverable: Draft Final Report & Slides – June 7

3.5 Phase 5: Finalization & Presentation

June 8 – June 28

The final phase encompasses the conclusive evaluation of the models and the preparation of the final deliverables and class presentations.

Tasks:
- Final model evaluation
- Prepare final presentation
- Final presentation in class (June 18–19)

- Final report writing and revisions
- Submit sparring group minutes

Deliverables:
- Final Report – June 28
- Sparring Group Minutes – June 28

4 Communication outline

This section describes the communication styles we are committed to use over the project.

4.1 Written communication

We will use two Teams in Microsoft Teams for our basic communication.

1. **DsLab26S_infrared.city_1**: This Team contains two channels for two different approaches. The **General** Channel is used as communication level with Kavita Surana (our connection to the Lecture and the WU). The other channel (**team intern**) is only accessible by the team itself and is our main chat for communication over the project.

2. **DsLab_infrared.city_1-coaches**: This Team is the basic for communication with the data coaches from infrared-city. We were asked to separate this from the main communication source, but still keep a Microsoft Teams Team for this. So we created a different Team.

4.2 Regular Meetings

Our meetings will be held via Microsoft Teams in most cases. It is possible that we are able to meet in person for one or another defined meeting. But regularly we plan to keep the communication remote or at least hybrid. We organized already two regular weekly meetings for basic communication.

1. **Regular Team Meeting**: On Thursdays 19:00 we will meet as a team remotely and talk about how to continue the work over the next week.

2. **Meeting with Data Coaches from infrared.city**: On Fridays 12:30 we scheduled a short regular meeting with the data coaches from infrared.city.

3. **Irregular Meetings with Kavita Surana**: In cases we need feedback for something from the WU site we schedule a meeting with her. To easily schedule the meeting we got a booking site for her time.

4.3 Documentation of Meetings

Because of the many meetings it is possible that one or another person is not able to participate every time. To still be able to communicate with absent people we want to write meeting notes and documentation of the meeting output after every meeting.

4.4 File and Code Repository

Every written result (documentation and project related) will be added to an already created GitHub repository. The repository is accessible by the team. See `https://github.com/nikmaster/wu_dslab_infrared-city`.

5 Team Roles

The main project team consists of 4 persons which are responsible for different focus points. But still commit to the project and in case one person gets stuck, the focus has to be to work together and find a way to tackle the issue.

Person	**Role**	**Tasks**
Dominik Oberhumer	Organisation of Team	Preparation of Tools, Reminding of Deliverables and Timeplan
Moritz Hörmansdorfer	Data Scientist	Machine Learning Training of Model
Nina Salnikow	Data Engineer	Fetching and Preparation of Data
Sara Klasová	Data Scientist	EDA (Exploratory Data Analysis)

Besides these persons, we also have additional support from people which are related to the project:

1. **Kavita Surana**: is our supervisor from WU. She will help us regarding infrastructure and gives the needed points for grading at the university.
2. **Oana Taut** and **Vasiliki Fragkia** are our data coaches from infrared.city. They will support us with different approaches and sources how we can use the data to achieve our goal.

6 Literature

Following additional papers are planned to be used as sources:

1. A review of individual tree crown detection and delineation from optical remote sensing images [9]
2. Integrating Traditional and Deep Learning Methods to Detect Tree Crowns in Satellite Images [10]
3. A Sentinel-2 machine learning dataset for tree species classification in Germany [11]
4. Object-based classification of urban plant species from very high-resolution satellite imagery [12]
5. Improving Urban Tree Species Classification with High Resolution Satellite Imagery and Machine Learning [13]

References

[1] European Commission, “Increasing tree coverage to 30% in european cities could reduce deaths linked to urban heat island effect,” 2023. Abgerufen: 2026-03-15.

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[3] Copernicus Programme, “Copernicus data space browser,” 2025. Abgerufen: 2026-03-15.

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[5] O. Pesek, L. Krisztian, M. Landa, M. Metz, and M. Neteler, “Convolutional neural networks for road surface classification on aerial imagery,” *PeerJ Computer Science*, vol. 10, p. e2571, 2024.

[6] A. I. Aramendia, “Decoding the u-net: A complete guide,” 2024. Abgerufen: 2026-03-15.

[7] V. S. F. Garnot and L. Landrieu, “Panoptic segmentation of satellite image time series with convolutional temporal attention networks,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 4872–4881, 2021.

[8] Wu SemSys, “Beam tutorial repository,” 2023. BEAM-instructions.pdf, accessed 2026-03-15.

[9] J. Zheng, S. Yuan, W. Li, H. Fu, and L. Yu, “A review of individual tree crown detection and delineation from optical remote sensing images,” 2023.

[10] O. Durgut, B. Kallfelz-Sirmacek, and C. Ünsalan, “Integrating traditional and deep learning methods to detect tree crowns in satellite images,” 2025.

[11] M. Freudenberg, S. Schnell, and P. Magdon, “A sentinel-2 machine learning dataset for tree species classification in germany,” *Earth System Science Data*, vol. 17, pp. 351–367, 2025.

[12] P. Sicard, F. Coulibaly, M. Lameiro, V. Araminiene, A. De Marco, B. Sorrentino, A. Anav, J. Manzini, Y. Hoshika, B. B. Moura, *et al.*, “Object-based classification of urban plant species from very high-resolution satellite imagery,” *Urban Forestry & Urban Greening*, vol. 81, p. 127866, 2023.

[13] R. Wenger, C. Bressant, L. Roettele, G. Forestier, and A. Puissant, “Improving urban tree species classification with high resolution satellite imagery and machine learning,” in *IEEE Conference Publication*, IEEE, 2024.